

Shahid Syed Ali

C CONTACTS

Selangor Semenyih, 43500 Semenyih,
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S SKILLS

- C programming: Beginner
- Python & NumPy Lib: Intermediate
- SolidWorks: Intermediate
- MATLAB: Intermediate
- LTSpice: Beginner
- Ladder Logic: Beginner
- Industrial Automation & Embedded Control Systems (Microcontrollers)

M RELEVANT MODULES

Power and Energy, University of Nottingham

Sep 2024 - Apr 2025

Analogue, Digital Electronics and Computing, University of Nottingham

Sep 2024 - May 2025

Statics and Dynamics, University of Nottingham

Sep 2024 - May 2025

Electrical Energy Conditioning and Control, University of Nottingham

Sep 2025 - May 2026

Electronic Processing Robotics, University of Nottingham

Sep 2025 - May 2026

Thermodynamics and Fluid Mechanics 1, University of Nottingham

Sep 2025 - May 2026

Modelling: Methods and Tools, University of Nottingham

Sep 2024 - May 2025

Design, Manufacture & Project, University of Nottingham

Sep 2025 - May 2026

Engineering Mathematics, University of Nottingham

Sep 2024 - May 2025

Applied Mechatronics Engineering Construction Project 1 &2, University of Nottingham

Sep 2024 - May 2026

REFERENCES

- 1) Dr Hermawan Nugroho
hermawan.nugroho@nottingham.edu.my
- 2) Ir. Ts. Nandan Ganesh Jayabalan
nandanganesh@nottingham.edu.my

Mechatronics Engineering Student

I PROFESSIONAL SUMMARY

Second-year Mechatronics Engineering student with a strong interest in robotics, embedded systems, and intelligent automation. Passionate about solving real-world problems by integrating mechanical design, electronics, and software, with hands-on experience in CAD and prototyping. Proactive, collaborative, and eager to contribute to innovative engineering projects while continuously developing technical and practical skills.

E EXPERIENCE

University of Nottingham

Selangor Semenyih

Industrial Conveyor Test Rig + Robot Arm Pick & Place

Sept 2025 - Apr 2026

- Implemented an Arduino Mega as the real-time controller for an ICT3/Bytronic peg-ring verification rig, using opto-isolated inputs and relay-driven outputs to interface 24 V industrial devices with 5 V logic. Produced wiring schematics, interface diagrams, and I/O maps, and supported commissioning/validation. Also integrated a Raspberry Pi + robot arm for pick-and-place using forward/inverse kinematics and motion sequencing; designed in SolidWorks, validated in simulation, and supported 3D-printed fabrication with awareness of PLC concepts, safety interlocks, and ladder sequencing.

University of Nottingham

Selangor Semenyih

Centrifugal Clutch Design & Manufacture

Sept 2025 - Apr 2026

- Contributed significantly to a group design-and-make project delivering a leading-leading centrifugal clutch meeting the test-rig requirement of ≥ 0.3 N·m at 2000 rpm with disengagement below 1000 rpm. Modelled key parts and the full assembly in SolidWorks, checked fit/clearances, and produced GA + BS 8888 manufacturing drawings. Verified performance through engineering calculations (centrifugal/spring forces, pressure, torque), prepared the BOM/cutting list and process sheets, and supported CNC fabrication, workshop fitting, assembly, and functional testing to ensure reliable engagement and alignment.

University of Nottingham

Selangor Semenyih

Pit Stop Challenge Pneumatic Mechanism

Feb 2025- Apr 2025

- Contributed significantly to a group and built a pneumatically actuated piston mechanism for the Pit Stop Challenge pick-and-place task (foam block transfer under time constraints). Modelled all parts in SolidWorks, created a fully constrained assembly, and checked motion/clearances before fabrication. Produced BS 8888 drawings, BOM/cutting list, and process sheets, then manufactured and assembled components using lathe work, drilling, cutting, deburring, and fitting to ensure accurate alignment and reliable operation on the pneumatic rig.

University of Nottingham

Selangor Semenyih

Advanced Line Following Robot Car

Sept 2024 - Dec 202

- Designed and built a 4WD Arduino line-following robot with a custom Veroboard IR sensor module for accurate black-line tracking. Integrated ultrasonic obstacle detection, rotary encoder distance measurement, and MPU6050 orientation/ramp sensing, with an LCD for live system data. Developed embedded control code for sensor fusion and motor control, added HC-05 Bluetooth mobile operation, and delivered a fully functional robot meeting all milestones.

E EDUCATION

BEng + MEng (Hons), University of Nottingham

Semenyih, Malaysia

First Year – 73% First Class Honors

Sept 2024- May 2028

Higher Secondary, Al Yasmin International School

Riyadh, Saudi Arabia

Score -90.2% (CBSE-India)

March 2023 – May 2024

L LINKS

<https://www.linkedin.com/in/shahid-syed-ali-284492376>